

MIET2115

Assignment 2

Material Selection for a Range of Applications

Group 29

Due Date: 09/06/2026

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Hugh Williams		Report Format, Introduction, Method, Reviewing and Altering other's sections, Bumper Bar for a Family Car Application, Window Latch {Component Design and Function, Candidate Materials} All {Operating Environment, Required Mechanical and Physical Properties, Tabulated Material Properties, Normalised Weighted Decision Matrix} Appendices, References
		Pressure Vessel {Component Design and Function, Candidate Materials, Final Material Selection}
		Liquid Hydrogen Tank {Component Design and Function, Candidate Materials, Final Material Selection}
		Reviewing and Altering other's sections, Window Latch {Final Material Selection}, Finding figures, Turbine Blade {Component Design & Function, Final Material Selection} Comparative Discussion, Conclusion
		Turbine Blade {Component Design and Function, Operating Environment Discussion, Required Mechanical and Physical Properties Discussion, Candidate Materials, Data Reliability and property Data Discussion, Weighted Decision Matrix Discussion, Final Material Selection}

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Executive Summary

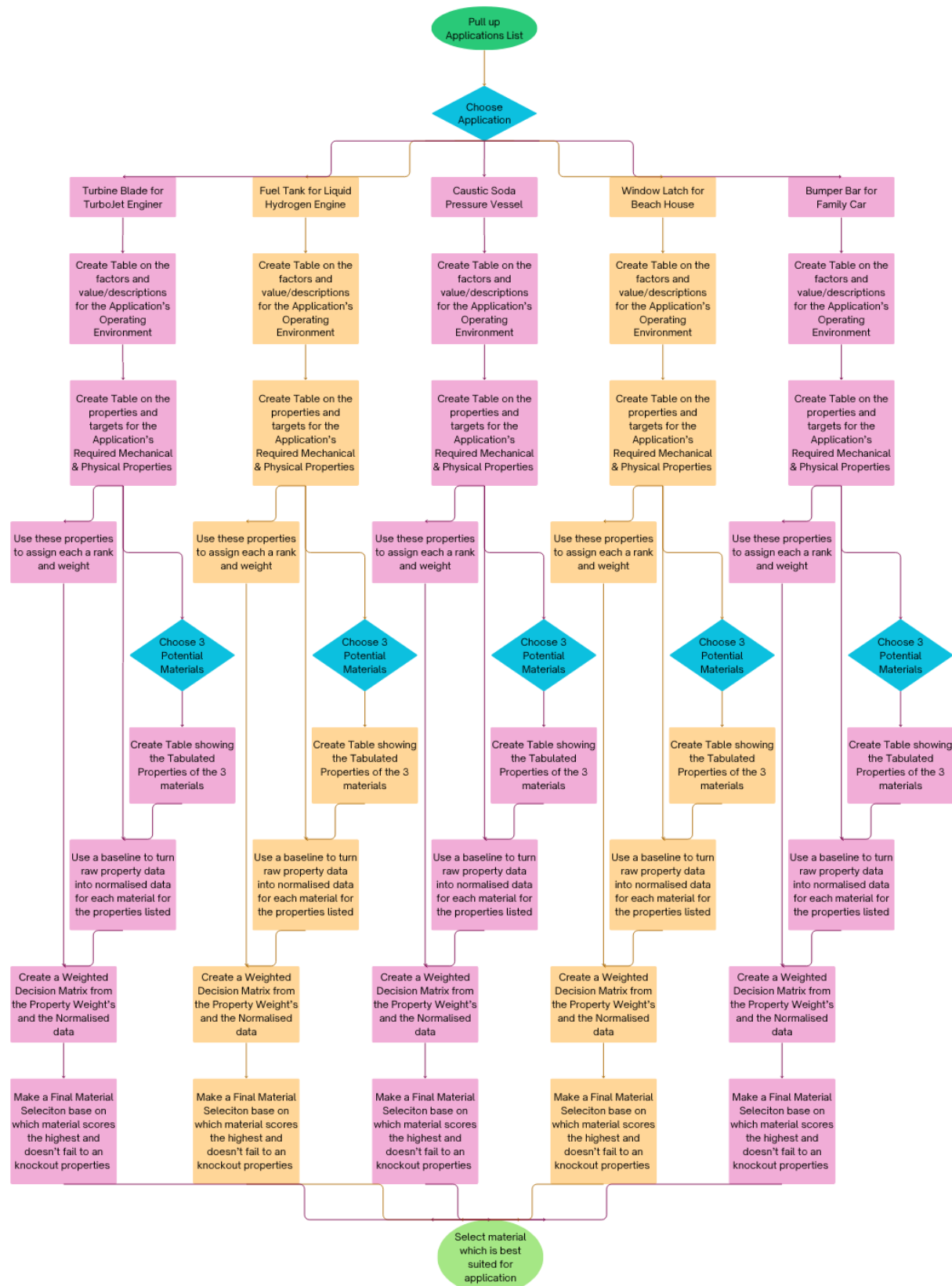
In this report there is a material selection process for five applications varying in environments and requirements. To complete this first each applications operating environment was found, then the required mechanical and physical properties were outlined and named knock-out properties, from this three candidate materials were selected where they were then put in a weighted decision matrix to find the best suited material. For the bumper bar the selected material was DP590 Dual-Phase Steel, for the window latch the chosen material was C95800 Nickel-Aluminium Bronze and Monel 400 was the best fit for the Caustic Soda Pressure Vessel. The material selected for the Liquid Hydrogen Tank was 316L Stainless Steel and for the Turbo Jet Engine Turbine Blade Rene 41 was the best fit. Across all applications performance was prioritised over weight and cost, which may need to be evaluated for weight sensitive applications such as the turbine blade, however there is still great confidence in all final materials as using the knockout criteria excluded unsuitable materials. Additionally, the weighted decision matrix method meant that material selected was the most suitable for its application.

Introduction

Material selection is an essential part of the design process where engineers compare different materials to find which is most suitable for the application. The correct material choice directly governs applications' safety, cost, lifespan and many more through its properties. For this report the five applications a material selection will be performed on are a bumper bar for a family car, a window latch for a holiday house on the beach, a pressure vessel containing caustic soda, a fuel tank for a liquid hydrogen engine, and a turbine blade for a turbojet engine. For each application, three materials will be selected for consideration, then by going through a selection criteria, one material will be chosen. For these applications we assume that the material selection is for one component and not uniform across the entire application. Furthermore, using reliable resources, informed assumptions must be made for a material's minimum criteria when operating in that application. Following the introduction, there will be a methodology which leads into the material selection process for the five applications and then a conclusion for what was found in the report.

Methodology

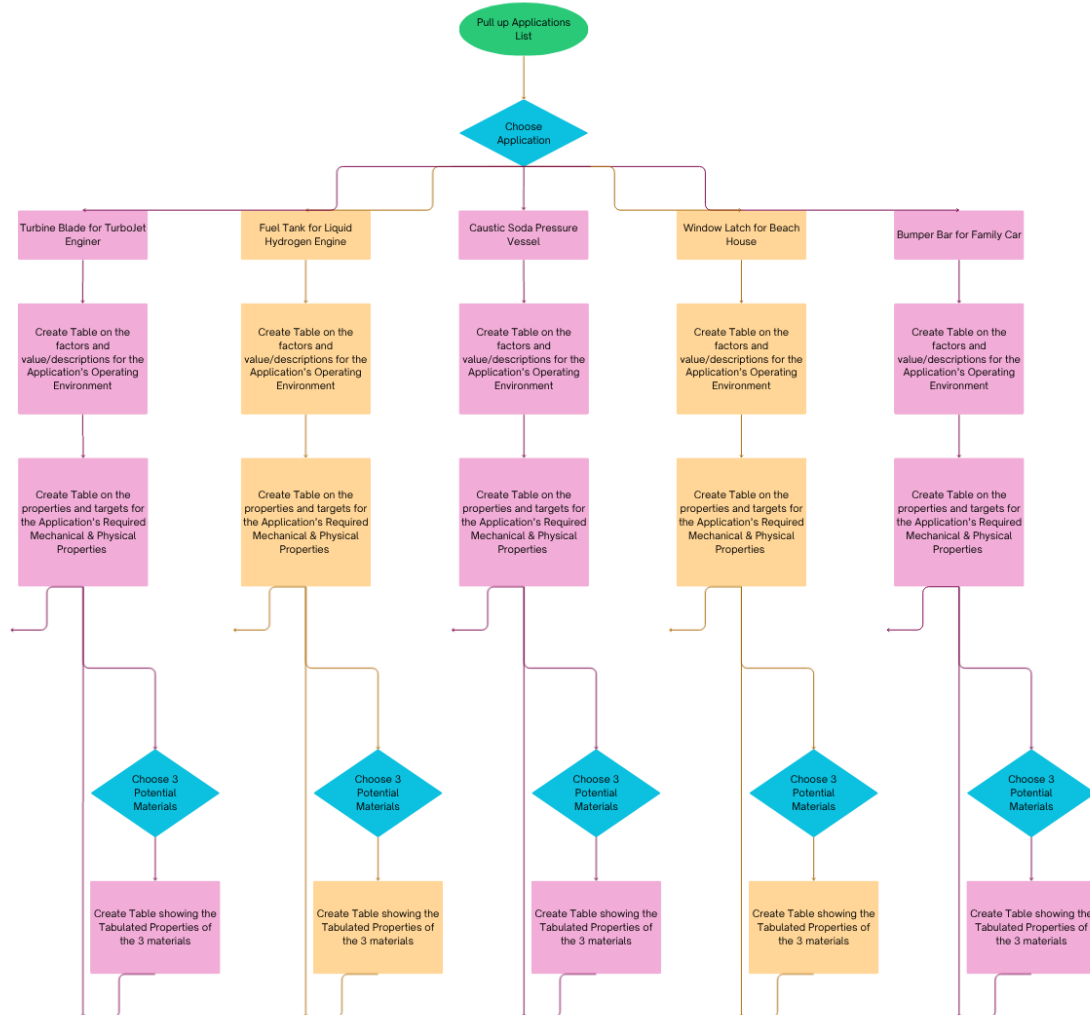
Figure 1, Material Selection Original Flow Chart



Selection Process

The selection process can be seen from this section of the methodology flow chart.

Figure 2, Material Selection Process extert from [Figure 1](#)

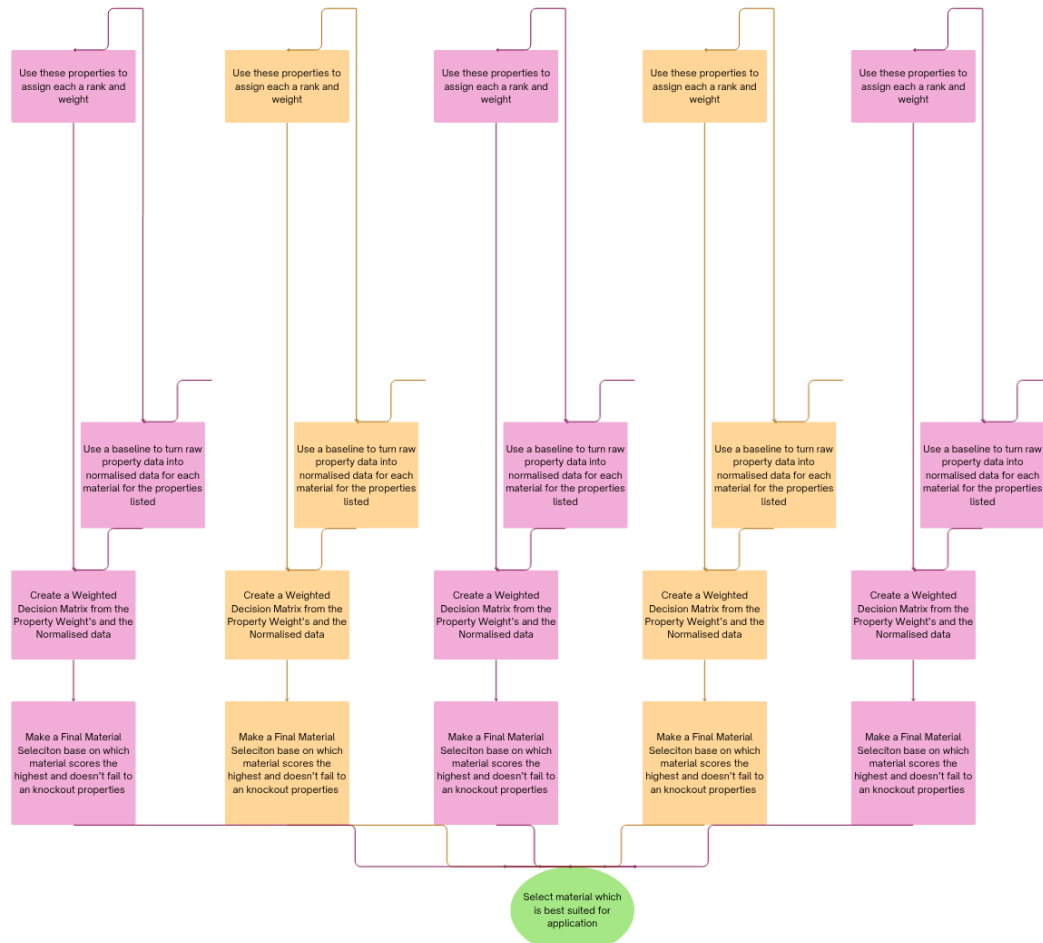


From this it shows how before selecting candidate materials for each application, engineers must know the operating environment and required mechanical and physical properties. If those specifications aren't known then a material may be misidentified to for selection as a candidate material.

Weighted Decision Matrix Method

For the weighted decision matrix, we can see which area of the flow chart it's found in.

Figure 3, Normalised Weighted Decision Matrix Flowchart exert from [Figure 1](#)



From the required mechanical and physical properties of each application, they must be given weights relative to how they rank on importance for the application, with certain values having to be capped if they are outliers. This equation shows how the weights are given relative to the properties' rankings.

Equation 1, Property Weighting

$$w_{property} = \frac{n_{properties} - R_{rank} + 1}{\frac{n_{properties}(n_{properties} + 1)}{2}}$$

To find the material best suited for an application first the materials' raw properties must be normalised against a baseline. This is done through the two equations:

Equation 2, Normalised Material Property for Properties where Higher is Better

$$n_{material} = \frac{x_{raw}}{x_{base}}$$

Equation 3, Normalised Material Property for Properties where Lower is Better

$$n_{material} = \frac{x_{base}}{x_{raw}}$$

The normalised material properties are then multiplied by the weighting for each relative property where the final score is given through the equation:

Equation 4, Final Material Score

$$\text{Score}_{material} = \sum(n_{property} \times w_{property})$$

It should also be noted that for the final material it must pass all the knockout property values to be selected. The knockout properties are selected as there are standards and minimum property values that the material must meet or not be available for selection.

Non-Technical Considerations

There are also non-technical considerations to be made for a material selection which impact the final score. These may include cost, recyclability and carbon footprint and are essential for the decision-making process as they can alter the materials' feasibility. These properties were given a lower weighting relative to the technical considerations due to them generally not being considered as essential and rather as additional properties.

Data Sources & Reliability

The quantity of data obtained for this report must be from quality sources such as datasheets and standards documents. If none are available then the next source type to use are handbooks and then to websites such as scientific blogs and markets for prices.

Discrepancies between data on different sites may appear. To decide on which data piece to apply in the report, the data point found closest to the operating conditions is to be used as the primary, if there are no differences between those then take the data from the most reputable source (using the method from the above paragraph).

Applications

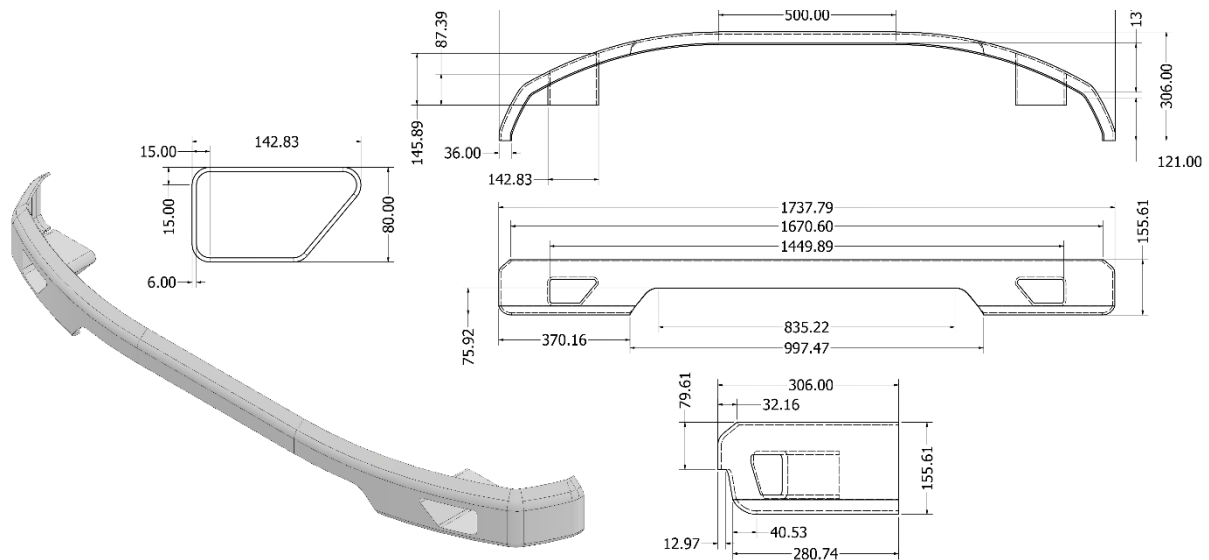
Bumper Bar for a Family Car

Component Design & Function

This section focuses on selecting the material which the reinforcement beam of a bumper bar will be made of. The reinforcement beam is the primary component for absorption and redistribution of energy in the result of a crash.

A bumper bar is a part of the car designed to protect the car body from any damage. A bumper bar must adhere to Australian Standard AS 4876.1 which includes requirements to minimise injury to pedestrians, match the vehicle profile, provide adequate strength, be securely attached and provide a controlled absorption of energy. Additionally, for use it must follow requirements set out in FMVSS Part 581 (USA), UN ECE R42 (EU) and IIHS bumper test protocol.

Figure 4, Impact Bar for Family Car Diagram



(Sugiri et al. 2025, fig. 1)

Reinforcement beam's cross-sectional area is an essential aspect for increasing beam strength and damping ability. This requires a material with high formability for a maximum section modulus. Torsional stiffness is directly proportional to the section height, preventing excessive deformation of the beam. The beam curvature should also follow the vehicles body shape as the curvature around the front of the car assists in distributing loads through the reinforcement beam. The thickness of the beam is generally between 2mm – 7mm with stronger materials such as steel being on the lower end and materials such as composites at the upper echelon, which assists in finding an equilibrium between strength and weight. Ribs are utilised for stiffness, distortion resistance and rigidity while minimising material use and for optimal use should mimic the load direction. The joint location and method impact energy absorption and load transfer through dictating the deformation sequence. The dimensions of the beam are variable to the target absorption as well as the vehicle specifications.

Operating Environment

Table 1, Bumper Bar Operating Environment

Factor	Value/ Description	Additional Info
Service Temp Range	-40°C-85°C	(ISO 2023)
Pressure	No Internal Pressure	N/A
Mechanical Loading	1738.89 Joules	Assuming weight of 1800kg at 5km/h crash $E = \frac{1}{2}mv^2$ (Cars24 Australia 2026)
Atmospheric Exposure	water, humidity, dust, UV, temperature change	Sits in the bumper bar but is still open to exposure
Chemical Environment	Salt, water, mud, oil, car fluids, cleaners	11-year useful life of potential corrosion ("40 CFR 86.1805-04 -- Useful life." n.d.)
Service Life	10-15 years or 150000-250000 km	("40 CFR 86.1805-04 -- Useful life." n.d.)
Failure Mode to Avoid	Plastic collapse before target energy, corrosion, buckling in wrong locations, joint failures, excess than planned obsolescence	(Du et al. 2023)
Assumed Manufacturing	heat treatment, cold welding	what makes the strongest material
Joining Method	welded, bolted, riveting	Which best fits the selected material
Maintenance	not need maintenance if no collisions occur	corrosion, cracking at joints
End of life	Best possible recyclability	Recyclability and carbon footprint

Required Mechanical and Physical Properties

Table 2, Bumper Bar Required Mechanical and Physical Properties

Required Property	Quantitative Target	Reason
Yield Strength (KO)	>220 MPa	resist deformation at crash speed
Density	Minimum	fuel economy
UTS	>330 MPa	Ultimate load capacity
Fracture Toughness	>35 MPa*m ^{1/2}	avoid brittle fracture at min service temp
Elongation at break	>10%	plastic strain for energy absorption
Fatigue Strength	>100 MPa	repeated loads of service life
Elasticity (KO)	>50 GPa	deflection during impact effecting crumple zone
CTE	<25*10 ⁶	thermal stress in joints
Service Temperature	85C	not degrade from conditions in service life
Corrosion Resistance	<0.05 mm/yr	servicing environment
Cost per kg	<5 AUD/kg	budget
Carbon Footprint per kg	<4.5 kgCO ₂ /kg	sustainability
Recyclability	>85% (high)	Sustainability

Candidate Materials

The candidate materials for a reinforcement beam of a bumper bar are DP590 Stainless steel, AA6061-T6 Aluminium and Thermoplastic Olefins. The metals were chosen as they have high strengths and the thermoplastic was chosen to be an alternate material which still sees use in bumper bars but can demonstrate how some properties are essential for a reinforcement beam.

Tabulated Material Properties

Table 3, Bumper Bar Tabulated Material Properties

Property	Units	DP590	AA6061-T6 Aluminium	Thermoplastic Olefins (TPO)
Density	g/cm ³	7.85	2.7	0.90-0.93
Young's Modulus	GPa	219	68.9	0.000986-2.71
Yield Strength	MPa	330-430	276	20
UTS	MPa	590-700	310	8-25
Fracture Toughness	MPa·m ^{0.5}	50-80	29	3.17
Elongation at Break	%	19-27	10	600
Fatigue Strength	MPa	354.2	96	N/A
Hardness	HB	121-183	93	Shore 60A-70D
CTE	µm/m·°C	15.4	23.6	68.7
Thermal Conductivity	W/m·K	50	167	0.11
Melting Point	°C	1425-1540	580	114
Specific Heat	J/kg·K	460	900	1700-2500
Service Temp Limit	°C	400	170	100
Corrosion Rate	mm/yr	0.045	0.051	not consistent
Cost	AUD/kg	1.20-1.80	5.17	3.49-6.97
Embodied Energy	MJ/kg	38.8	150	75-83
Carbon Footprint	kgCO ₂ /kg	2.18	8.3	2.9-3.2
Recyclability	--	High	Infinite	5-6 rating
Machineability	--	Moderate	Good	Excellent
Poisson's ratio	--	0.3	0.33	0.45-0.48

Derived from (ASM International 2005), ("6061-T6 Aluminum :: MakeltFrom.com" 2020), ("Aluminum 6061-T6; 6061-T651" n.d.), ("DP 590 Steel: High-Strength Automotive grade material" 2026), ("Overview of materials for Thermoplastic Olefinic Elastomer (TPO)" n.d.), ("Plastic Thermal Conductivity Reference Table - Alfa Chemistry" 2024), ("SAE J2020 Accelerated Weathering Testing for Automotive Materials" n.d.), (Api 2025, sec. 3.2, 3.3, 4), (autoplasticsworld.com 2025), (Elsevier Inc. 2013), (Feng et al. 2025), (Gabrian International (H.K.) Ltd. n.d.), (Hellier, Chaphalkar & Prusty 2017, sec. 2.1), (International Aluminium Institute 2024), (Kg n.d.), (Khatibpour 2025), (Kloeckner Metals 2026), (Kruse Training, Inc. n.d.), (Luo, Li & Xing 2022, sec. Abstract), (LyondellBasell 2019), (Ma et al. 2017), (Mechanical behaviour of engineering materials 2007), (Nikhare, Asgari & Weiss 2009), (Sung, Kim & Wagoner 2010, sec. 3.7), (worldsteel 2026), (Xie et al. 2017), (Xu & Juang 2021, sec. 3.4), (Yasin, Terry & Taylor 2023b, sec. 4.6.3), (Your Home 2020), (Zhang et al. 2019, sec. 4.4).

Normalised Weighted Decision Matrix

For Yield Strength (MPa):

Equation 5, Bumper Bar Normalising Yield Strength for DP590

$$n_{DP590} = \frac{\sigma_{raw}}{\sigma_{base}} = \frac{380}{220} = 1.73$$

Equation 6, Bumper Bar Normalising Yield Strength for AA6061-T6

$$n_{AA6061-T6} = \frac{\sigma_{raw}}{\sigma_{base}} = \frac{276}{220} = 1.25$$

Equation 7, Bumper Bar Normalising Yield Strength for TPO

$$n_{TPO} = \frac{\sigma_{raw}}{\sigma_{base}} = \frac{20}{220} = 0.09$$

Table 4, Bumper Bar Weight Allocation

Rank	Property	Weighting	Justification
3	Yield Strength	0.12	Main structural property (KO)
1	Density	0.14	Lower weight
5	UTS	0.10	Max load for higher energy impacts
4	Fracture Toughness	0.11	prevention of failure mode
6	Elongation at break	0.09	energy absorption
8	Fatigue Strength	0.07	strength over time
7	Elasticity	0.08	controls deflection (KO)
12	CTE	0.02	Prevent warping
10	Service Temperature	0.04	If it can operate in conditions
9	Corrosion	0.05	Not break down over time
2	Cost per kg	0.13	Lowest cost
11	Carbon Footprint	0.03	Sustainability
13	Recyclability	0.01	End of life commitment
	Σ	1.00	

Table 5, Bumper Bar Normalised Material Properties

Property	Baseline	DP590 raw	DP590 norm	AA6061-T6 raw	AA6061-T6 norm	Thermo plastic raw	Thermo plastic norm
Yield Strength (MPa)	220	380	1.73	276	1.25	20	0.09
Density (g/cm ³)	7.85	7.85	1.00	2.7	2.91	0.92	3.00
UTS (MPa)	330	645	1.95	310	0.94	16.5	0.05
Fracture Toughness (MPa·m ^{0.5})	35	65	1.86	29	0.83	3.17	0.09
Elongation at break (%)	10	23	2.30	10	1.00	600	3.00
Fatigue Strength (MPa)	100	354	3.00	96	0.96	0	0.00
Elasticity (GPa)	50	219	3.00	68.9	1.38	1.35	0.03
CTE (µm/m·°C)	25	15.4	0.62	23.6	0.94	68.7	2.75
Service Temperature (°C)	85	400	3.00	170	2.00	100	1.18
Corrosion (mm/yr)	0.05	0.045	1.11	0.051	0.98	0.01	3.00
Cost per kg (AUD/kg)	5	1.5	3.00	5.17	0.97	5.23	0.96
Carbon Footprint (kgCO ₂ /kg)	4.5	2.18	2.06	8.3	0.54	3.05	1.48
Recyclability (%)	85	85	1.00	100	1.18	55	0.65
(colour means capped at 3)							

Table 6, Bumper Bar Weighted Decision Matrix

Property	Weight	DP590 norm	DP590 ×W	AA6061-T6 norm	AA6061-T6×W	Thermo plastic norm	Thermo plastic ×W
Yield Strength (MPa)	0.12	1.73	0.2088	1.25	0.1516	0.09	0.0110
Density (g/cm ³)	0.14	1.00	0.1429	2.91	0.4153	3.00	0.4286
UTS (MPa)	0.10	1.95	0.1933	0.94	0.0929	0.05	0.0049
Fracture Toughness (MPa·m ^{0.5})	0.11	1.86	0.2041	0.83	0.0911	0.09	0.0100
Elongation at break (%)	0.09	2.30	0.2022	1.00	0.0879	3.00	0.2637
Fatigue Strength (MPa)	0.07	3.00	0.1978	0.96	0.0633	0.00	0.0000
Elasticity (GPa)	0.08	3.00	0.2308	1.38	0.1060	0.03	0.0021
CTE (μm/m·°C)	0.02	0.62	0.0135	0.94	0.0207	2.75	0.0604
Service Temperature (°C)	0.04	3.00	0.1319	2.00	0.0879	1.18	0.0517
Corrosion (mm/yr)	0.05	1.11	0.0611	0.98	0.0539	3.00	0.1648
Cost per kg (AUD/kg)	0.13	3.00	0.3956	0.97	0.1275	0.96	0.1261
Carbon Footprint (kgCO ₂ /kg)	0.03	2.06	0.0681	0.54	0.0179	1.48	0.0486
Recyclability (%)	0.01	1.00	0.0110	1.18	0.0129	0.65	0.0071
Σ			2.061		1.329		1.179

Table 7, Bumper Bar Final Material Selection

Rank	Material	Σ	Knock out check
1	DP590 Stainless Steel	2.061	✓
2	AA6061-T6 Aluminium	1.329	✓
3	Thermoplastic Olefin	1.179	X - Fails to meet required Yield Strength and Elasticity

Final Material Selection

The selected material for use in the reinforcement beam is DP590 Dual phase Steel, hot rolled, annealed.

Whilst DP590 Dual phase steel scored the highest of the three materials it does come with some trade-offs. The steel has a density of 7.85 g/cm^3 compared to AA6061-T6 of 2.7 g/cm^3 and TPO of 0.92 g/cm^3 , so the beam's mass will be greater, reducing the family car's fuel efficiency. Additionally, at the end of service life AA6061-T6 is infinitely recyclable, whereas DP590's recyclability is limited. Furthermore, TPO has a far smaller rate of corrosion (0.01 mm/year) than DP590 (0.045 mm/year) meaning that theoretically TPO can sustain greater periods without servicing.

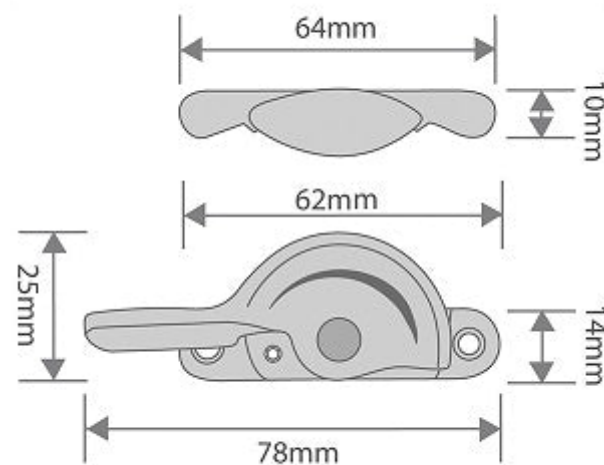
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Window Latches for a Holiday House on the Beach

Component Design & Function

A window latch is split into two parts, one sitting on the window frame and the other on the window. When the window is closed a latch is engineered to keep the window from opening without a user opening it. The latch operates by moving an arm into a fixed groove preventing movement in the direction of the hinges.

Figure 5, Window Latch Diagram



(Sellaseal n.d.)

A window latch at a holiday house must be resistant galvanic corrosion as the proximity to the beach will increase exposure to saltwater. Additionally, as it is a holiday house the engineers must assume the house will be empty for extended periods, which increases the risk of burglary. So, the latch must also be strong enough to prevent intruders from breaking the arm of and opening the window for easy access. Furthermore, mounting points concentrate stress and may be a failure point under excessive force if located in certain regions. The arm dimensions must also be considered as longer arms will fracture faster relative to shorter arms due to the applied load producing a greater bending moment, however increasing the thickness is a method to combat this. The cross-sectional area of the window frame will impact how the arm secures the window, determining the necessary latch design to optimise the gasket compression, which is also essential for blocking environmental conditions from entering the residence.

Operating Environment

Table 8, Window Latch Operating Environment

Factor	Value/ Description	Additional Info
Corrosion category	ISO 9223 category C5/CX	Costal atmospheric conditions (ISO 2012)
Chloride source	salt spray, marine aerosols	ocean excrement accumulation
Service Temp	-5°C-55°C	excess on maximum temps of a year
Humidity	condensation	salt and humidity increase corrosion risk
UV	direct sunlight	durable under continuous UV rays
Mechanical loading	>400N force	latch holding as well as during forces applied by potential burglary (Das & Wang 2004)
Galvanic compatibility (KO)	Aluminium window frame	not accelerate corrosion
Service Life	15-20 years	not need frequent replacement
Failure modes to avoid	pitting, corrosion, seized mechanism, excess force	corrosion and purposeful breaking driven
Manufacturing	Coated to prevent corrosion	leads to failures
End of life	Recyclable	minimise waste

Required Mechanical and Physical Properties

Table 9, Window Latch Required Mechanical and Physical Properties

Required Property	Quantitative Target	Reason
Corrosion rate	minimum possible	greatest risk to failure
Galvanic compatibility	not accelerate window frame	prevent acceleration of corrosion
Yield strength	>150 MPa	Withstand excessive force as well as force applied in everyday use
Fatigue Strength	>65 MPa	not weaken under continuous use over large periods of time
Hardness	medium-high	not buckle under stress and prevent scratching
Density	low	lighter weight improves usability
Cost per kg	low	cheapest product
Carbon Footprint per kg	low	sustainability
Recyclability	high	sustainability
Manufacturability	medium-high	maximise speed whilst also forming the correct geometry

Candidate Materials

The candidates most suited for these operating environments are 316L Stainless Steel, C95800 Nickel-Aluminium Bronze, and AA6063 Anodised Aluminium. These are all metals and are credible candidates for this as they are used and tested materials suitable for outdoor use. They all have quite high yield strengths and fatigue strengths as well as a low corrosion rate.

Tabulated Material Properties

Table 10, Window Latch Tabulated Material Properties

Property	Units	316L Stainless Steel	Nickel Aluminium Bronze cast UNS C95800	AA6063 Anodised Aluminium
Density	g/cm ³	8.03	7.64	2.7
Young's Modulus	GPa	193	114	69.5
Yield Strength	MPa	255.11	241	170 (proof stress)
UTS	MPa	530.9	586	215
Fracture Toughness	MPa·m ^{0.5}	268	125	19.62
Elongation at Break	%	55	18	12
Fatigue Strength	MPa	270	210	68.9
Hardness	HB	139	159	75
CTE	µm/m·°C	16	15.5	23.5
Thermal Conductivity	W/m·K	12.1	36	201
Melting Point	°C	1390-1440	1043	655
Specific Heat	J/kg·K	500	440	900
Service Temp Limit	°C	925	-190-325	-273-(130-150)
Corrosion Rate	mm/yr	0.0045 (high temps) - 0.0019 (low temps)	Excellent (0.025-0.076 seawater)	0.024
Cost	AUD/kg	3.98	11.73	3.21
Embodied Energy	MJ/kg	80	33	190-210
Carbon Footprint	kgCO ₂ /kg	12	3.3	12.5-13.8
Recyclability	--	1	>50%	76
Machineability	--	40% (Moderate)	20	0.5
Poisson's ratio	--	0.27-0.3	0.32	0.33
Galvanic Compatibility with Aluminium	--	No	Yes	Yes

Derived from (Alloys n.d.), ("AISI Type 316L" n.d.), ("Aluminium alloy - Commercial alloy - 6063 - T6 extrusions" n.d.), ("Aluminum 6063-T6" n.d.), ("C95800 Alloy, C958 Nickel-Aluminum bronze - Wieland Concast" n.d.), ("Fatigue properties and endurance limits of stainless steels – British Stainless Steel Association" n.d.), ("FROM CAN TO SHEET: A CRITICAL REVIEW OF ALUMINUM CAN RECYCLING PROCESSES, TECHNOLOGIES, AND SUSTAINABILITY DIMENSIONS (2020–2025)" n.d., sec. 3.2), ("Machining DataSheet for Stainless Steel 316 (X5CrNiMo17-12-2)" n.d.), ("Metal Compatibility | Galvanic Corrosion Scale" n.d.), ("NICKEL ALUMINIUM BRONZE - Aluminium Bronze Price Per Kg Trader - Retailer from Mumbai" n.d.), ("Rectangular Aluminum 6063 T6 Block" n.d.), ("Stainless Steel (316) Price Trend, Index and Forecast" n.d.), (Ahmad & Enemuoh 2020, fig. Equation 5, Table 2), (AK Steel Corporation 2007), (AMERICAN IRON AND STEEL INSTITUTE 2020, fig. Table 5), (ASM Handbook, Volume 19: Fatigue and Fracture 1996,

p. 1814), (Barnby et al. 1977, sec. Abstract), (CORROSION BEHAVIOUR OF ALUMINUM ALLOY IN SEAWATER 2010, fig. 4), (Graedel et al. 2011, sec. Summary), (Li et al. 2021, sec. 3.3), (Lion 2025, sec. 3.2), (Liu et al. 2022, sec. 3.1), (Pandouria et al. 2021, fig. Table 3.1, sec. 3.4.1), (Products 2024), (Properties of Aluminum 6063 T5 n.d.), (Richardson & Copper Development Association 2016, fig. Table 9a, sec. 6.1.1, sec. 5.2), (World Material n.d.).

Normalised Weighted Decision Matrix

For Yield Strength (MPa):

Equation 8, Window Latch Normalising Yield Strength for 316L

$$n_{316L} = \frac{\sigma_{raw}}{\sigma_{base}} = \frac{255.11}{255.11} = 1.00$$

Equation 9, Window Latch Normalising Yield Strength for C95800

$$n_{C95800} = \frac{\sigma_{raw}}{\sigma_{base}} = \frac{241}{255.11} = 0.94$$

Equation 10, Window Latch Normalising Yield Strength for AA6063

$$n_{AA6063} = \frac{\sigma_{raw}}{\sigma_{base}} = \frac{170}{255.11} = 0.67$$

Table 11, Window Latch Weight Allocation

Rank	Property	Weighting	Justification
1	Corrosion rate	0.20	Most common failure mode
2	Yield strength	0.18	resists load from potential break ins
3	Fatigue Strength	0.16	repeated use of latch
4	Hardness	0.13	wear of latch
7	Density	0.07	ease of use
5	Cost per kg	0.11	purchase price of latch from stores
8	Carbon Footprint per kg	0.04	sustainability
9	Recyclability	0.02	disposal considerations
6	Manufacturability	0.09	production feasibility
	Σ	1.00	

Table 12, Window Latch Normalised Material Properties

Property	Baseline	316L raw	316L norm	C95800 raw	C95800 norm	AA6063 Anodised raw	AA6063 Anodised norm
Corrosion rate	0.0019	0.0019	1.00	0.051	0.04	0.024	0.08
Yield strength	255.11	255.11	1.00	241	0.94	170	0.67
Fatigue Strength	270	270	1.00	210	0.78	68.9	0.26
Hardness	139	139	1.00	159	1.14	75	0.54
Density	8.03	8.03	1.00	7.64	1.05	2.7	2.97
Cost per kg	3.98	3.98	1.00	11.73	0.34	3.21	1.24
Carbon Footprint per kg	12	12	1.00	3.3	3.64	13.15	0.91
Recyclability	10	10	1.00	50	5.00	76	5.00
Manufactura bility	30	30	1.00	20	0.67	50	1.67
(colour means capped at 5)							

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Table 13, Window Latch Weighted Decision Matrix

Property	Weight	316L norm	316L×W	C95800 norm	C95800×W	AA6063 Anodised norm	AA6063 Anodised×W
Corrosion rate	0.20	1.00	0.2000	0.04	0.0075	0.08	0.0158
Yield strength	0.18	1.00	0.1778	0.94	0.1679	0.67	0.1185
Fatigue Strength	0.16	1.00	0.1556	0.78	0.1210	0.26	0.0397
Hardness	0.13	1.00	0.1333	1.14	0.1525	0.54	0.0719
Density	0.07	1.00	0.0667	1.05	0.0701	2.97	0.1983
Cost per kg	0.11	1.00	0.1111	0.34	0.0377	1.24	0.1378
Carbon Footprint per kg	0.04	1.00	0.0444	3.64	0.1616	0.91	0.0406
Recyclability	0.02	1.00	0.0222	5.00	0.1111	5.00	0.1111
Manufacturability	0.09	1.00	0.0889	0.67	0.0593	1.67	0.1481
Σ			1.000		0.889		0.882

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Table 14, Window Latch Final Material Selection

Rank	Material	Σ	Knock out check
1	C95800	0.889	✓
2	AA6063 Anodised	0.882	✓
3	316L Stainless Steel	1.000	X - Isn't Galvanically compatible with aluminium frame

Final Material Selection

AA6063 Anodised was chosen over C95800 despite having a lower score as it has the edge in properties that are mainly looked at in manufacturing, like cost per kg and recyclability. C95800 has stronger physical properties, such as fatigue strength and hardness, lasting longer and under more pressure load. However, it costs significantly more to manufacture than AA6063 Anodised which would be unfavourable when deciding on similar materials since the application is of high supply. AA6063 is also more environmentally friendly with a lower carbon footprint which is favourable for a largely supplied application.

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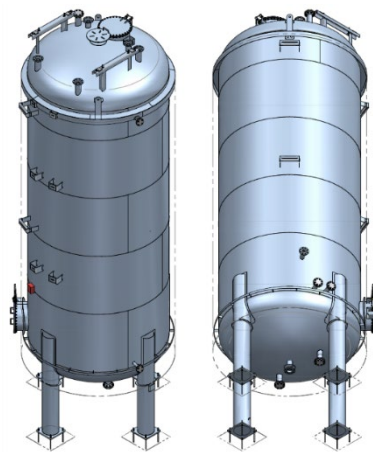
Pressure Vessel Containing Caustic Soda at 280 Degrees Celsius at 3.5 MPa

Component Design & Function

Caustic soda is a versatile, strong alkaline compound which is used the foundation material for many products from everyday cleaners to industrial products. Caustic soda goes by many names, including Sodium Hydroxide or Lye, and has the chemical composition NaOH.

Caustic soda is notably a highly corrosive compound which provides difficulties in construction materials for a storage tank. To design a pressure vessel to store caustic soda at 280° Celsius and 3.5MPa, the selection of a material is essential with no room for error as the possibility of leaks of the Caustic Soda could prove catastrophic due to its nature. The material must not only be able to withstand the 3.5MPa Pressure and 280° Celsius temperature, but also the extremely corrosive nature of the compound. Additionally, the Caustic Soda is presumed to be stored in an aqueous solution, this further produces difficulties in storage as any small cracks or corrosion in the vessel would have the solution escape easily.

Figure 6, Caustic Soda Pressure Vessel Diagram



(Balk 2025)

Pressure Vessels containing Caustic Soda are generally large cylindrical objects constructed out of metals. Important properties to consider for the material are the Caustic Resistance, Tensile strength, Yield strength, corrosion rate, cost, recyclability and Service temperature. Additionally, the weldability of the material is an essential component which can be overlooked as the vessels will have fittings welded for a variety of purposes. Welding these fittings on provides a strong connection that can keep the pressure inside of the vessel.

Operating Environment

Table 15, Caustic Soda Pressure Vessel Operating Conditions

Factor	Value/ Description	Additional Info
Internal Pressure	>3.5 Mpa	Given + (2023 ASME Boiler and Pressure vessel Code: SECTION VIII Rules for construction of pressure vessels Division 1 2023, secs. 35–6)
Service Temperature	280°C continuous	Given + (Bairamov 2011b)
Process fluid	50% NaOH	Given
Design Pressure	>3.85 Mpa	FOS
Design Temp	>295°C	FOS
PH	14	("pH and alkalinity" n.d.)
Stress	Hoop stress/cyclic loading	points of failure (cracking)
lifespan	20-30 years	industry average
Failure Modes	Caustic SCC, corrosion	caustic corrosion is the primary failure
Manufacturing	plates (rolled and welded), heat treatment	Industry specifications and standards (ASME BPVC)
Joining	Welding	ASME BPVC

Required Mechanical and Physical Properties

Table 16, Caustic Soda Pressure Vessel Required Mechanical and Physical Properties

Required Property	Quantitative Target	Reason
Caustic resistance (KO)	Safe	Predominant failure mode
standards allowed material (KO)	Allowed	(ASME Boiler and Pressure Vessel Committee on Pressure Vessels 2019, fig. Table UCS-23. Table UHA-23)
Allowable Design Tensile Stress (KO)	((SA-240)316L - 247 Mpa)	(ASME Boiler and Pressure Vessel Committee on Pressure Vessels 2019, fig. Table 44-4-1)
Weldability (KO)	Up to regulations	Weldable to remain pressurised
Yield Strength	Highest	Wall thickness
Corrosion Rate	Lowest	Thinning of wall over life
Cost	Lowest	Save money
Recyclability	Highest	End of life considerations
Service Temp Limit (KO)	>300°C	Able to continuously operate at operation temps
Embodied Energy	Lowest	Sustainability

Candidate Materials

316 Stainless steel is the second most widely used stainless steel behind 304 Stainless Steel [Austral Wright Metals, 2025]. 316 is known for its high corrosion resistance when compared to other steels as it contains 2% molybdenum which improves its overall resistance. 316 is typically used in marine applications due to its corrosive resistance and is seen in areas including Boat fittings, costal construction and more. 316 is also widely used in food processing and production, chemical processing, heat exchangers and more due to its corrosion resistance and strength.

Monel-400

Monel-400 is a nickel-copper alloy praised for its good strength, toughness and its excellent corrosion resistance. Widely used in marine applications, Monel-400 is known for its corrosion resistance, with many types of acids and nearly all alkalis. This alkali resistance is an essential

quality required for the pressure vessel. Additionally, Monel-400 is easily worked and fabricated with it also being able to be welded. Furthermore, the AS1210 and ASME Boiler and Pressure Vessel Code qualify Monel-400 for use in pressure vessels up to 475° Celsius [Austal Write Metals, n.d.]

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Tabulated Material Properties

Table 17, Caustic Soda Pressure Vessel Tabulated Material Properties

Property	Units	SA-516 Gr.70	316L Stainless Steel	Monel 400
Density	g/cm3	7.8	8.03	8.8
Young's Modulus	GPa	200	193	179.26
Yield Strength	MPa	363 at 300C	255.11	193-345 annealed
UTS	MPa	540 at 300C	530.9	482-586 annealed
Fracture Toughness	MPa·m ^{0.5}	98 kN/m at 300C	268	1260.4 kJ/m ²
Elongation at Break	%	31 at 300C	55	50-35 annealed
Fatigue Strength	MPa	250	270	21.0 annealed
Hardness	HB	140-200	139	110-140 annealed
CTE	µm/m·°C	12	16	15.7 at 300C
Thermal Conductivity	W/m·K	52	12.1	30.1 at 300C
Melting Point	°C	1510	1390-1440	1300-1350
Specific Heat	J/kg·K	470	500	470 at 300C
Service Temp Limit	°C	-46-538 (pressures of 50 Mpa)	925	537.78
Corrosion Rate	mm/yr	0.0088±0.0006	0.0045 (high temps) - 0.0019 (low temps)	0.03
Cost	AUD/kg	0.84-1.47	3.98	61.42
Embodied Energy	MJ/kg	48.4	80	110
Carbon Footprint	kgCO ₂ /kg	3.19	12	7.9
Recyclability	--	0.85	1	High
Machineability	--	55-80%	40% (Moderate)	satisfactory
Poisson's ratio	--	0.29	0.27-0.3	0.32
Caustic Resistance	mm/a	Poor (0)	poor (0)	0.00064 (100)

Derived from ("ASTM A516 Carbon Steel, Grade 70" n.d.), ("AISI Type 316L" n.d.), ("ASME SA 516 gr 70 Hardness Values_ Gangsteel" n.d.), ("ASTM A516 GR 70 Boiler Plate price in China market 2025-10_ Gangsteel" n.d.), ("Cold worked Monel 400 :: MakeltFrom.com" 2020), ("Fatigue properties and endurance limits of stainless steels – British Stainless Steel Association" n.d.), ("Machining DataSheet for Stainless Steel 316 (X5CrNiMo17-12-2)" n.d.), ("Machining DataSheet for Steel A516" n.d.), ("Monel 400 Scrap: Meaning and Uses?" n.d.), ("Stainless Steel (316) Price Trend, Index and Forecast" n.d.), ("What is temperature range of ASME SA 516 GR 70 material?_ Gangsteel" n.d.), (Ahmad & Enemuoh 2020, fig. Equation 5, fig. Table 2), (AK Steel Corporation 2007), (Alloys n.d.), (AMERICAN IRON AND STEEL INSTITUTE 2020, fig. Table 5), (ASM Handbook, Volume 19: Fatigue and Fracture 1996, p. 1814), (Grosu et al. 2018, sec. Results and discussion), (Hammond et al. 2008, p. 16), (Li et al. 2021, sec. 3.3), (Nickel Development Institute n.d., fig. Table 20), (Products 2024), (Renine Metalloys 2026), (Seok 2000, fig. Table 2, fig. Table 3), (Shanghai Shenghonghe Import and Export Co., Ltd 2025), (Special Metals Corporation 2005, fig. Table 2, fig. Table

3, fig. Table 4, fig. Table 15, fig. Table 19, sec. Creep and Rupture Properties, sec. Machining), (Steel 2024), (The Specialty Steel Industry of North America (SSINA) 2018, fig. Table 7), (Toyo 2025), (World Material n.d.), (World Steel Association 2021, p. 17), (Yadav, Misra & Tyagi 2024, sec. Conclusion).

Normalised Weighted Decision Matrix

For Yield Strength (MPa):

Equation 11, Caustic Soda Pressure Vessel Normalising Yield Strength for SA-516

$$n_{SA-516} = \frac{\sigma_{raw}}{\sigma_{base}} = \frac{363}{255.11} = 1.42$$

Equation 12, Caustic Soda Pressure Vessel Normalising Yield Strength for 316L

$$n_{316L} = \frac{\sigma_{raw}}{\sigma_{base}} = \frac{255.11}{255.11} = 1.00$$

Equation 13, Caustic Soda Pressure Vessel Normalising Yield Strength for Monel 400

$$n_{Monel\ 400} = \frac{\sigma_{raw}}{\sigma_{base}} = \frac{621}{255.11} = 2.43$$

Table 18, Caustic Soda Pressure Vessel Weight Allocation

Rank	Property	Weighting	Justification
1	Caustic resistance (KO)	0.22	Main failure mode
2	Allowable Design Tensile Stress (KO)	0.19	Hoop stress with design
3	Yield Strength	0.17	impacts wall thickness
4	Corrosion Rate	0.14	wall thinning
7	Cost	0.06	price of product
5	Recyclability	0.11	end of life considerations
6	Service Temp Limit (KO)	0.08	Able to operate continuously at temp
8	Embodied Energy	0.03	Sustainability
	Σ	1.00	

Table 19, Caustic Soda Pressure Vessel Normalised Material Properties

Property	Baseline	SA-516 raw	SA-516 norm	316L raw	316L norm	Monel 400 raw	Monel 400 norm
Caustic resistance (KO)	100	0	0.00	0	0.00	100	1.00
Allowable Design Tensile Stress (KO)	250	540	2.16	530.9	2.12	690	2.76
Yield Strength	255.11	363	1.42	255.11	1.00	621	2.43
Corrosion Rate	0.0045	0.0088	0.51	0.0045	1.00	0.03	0.15
Cost	3.98	1.47	2.71	3.98	1.00	61.42	0.06
Recyclability	100	85	0.85	100	1.00	85	0.85
Service Temp Limit (KO)	925	538	0.58	925	1.00	537.78	0.58
Embodied Energy	80	48.4	1.65	80	1.00	110	0.73
(colour means capped at 3)							

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Table 20, Caustic Soda Pressure Vessel Weighted Decision Matrix

Property	Weight	SA-516 norm	SA-516×W	316L norm	316L×W	Monel 400 norm	Monel 400×W
Caustic resistance (KO)	0.22	0.00	0.0000	0.00	0.0000	1.00	0.2222
Allowable Design Tensile Stress (KO)	0.19	2.16	0.4200	2.12	0.4129	2.76	0.5367
Yield Strength	0.17	1.42	0.2372	1.00	0.1667	2.43	0.4057
Corrosion Rate	0.14	0.51	0.0710	1.00	0.1389	0.15	0.0208
Cost	0.06	2.71	0.1504	1.00	0.0556	0.06	0.0036
Recyclability	0.11	0.85	0.0944	1.00	0.1111	0.85	0.0944
Service Temp Limit (KO)	0.08	0.58	0.0485	1.00	0.0833	0.58	0.0484
Embodied Energy	0.03	1.65	0.0459	1.00	0.0278	0.73	0.0202
Σ			1.067		0.996		1.352

Table 21, Caustic Soda Pressure Vessel Final Material Selection

Rank	Material	Σ	Knock out check
1	Monel 400	1.352	✓
2	SA-516	1.067	X- Not high enough caustic resistance
3	316L Stainless Steel	0.996	X- Not high enough caustic resistance

Final Material Selection

The chosen material to create the pressure vessel out of is Monel 400. The nickel-copper alloy was chosen as not only did it score the highest with the weighted decision matrix, but it was also the only material that had a high enough corrosion resistance. As Caustic Soda is a highly corrosive material, the corrosion resistance of the material chosen to store it is an essential property. Although both 316-Stainless Steel and SA-516 are considered corrosion resistant materials, the extreme alkalinity of the Caustic Soda solution would still prove to corrode the materials particularly at the high temperature and pressure. This would lead to the pressure vessel becoming damaged and needing replacement. Monel 400 however, is renowned for its extreme corrosion resistance due to its composition of nickel and copper.

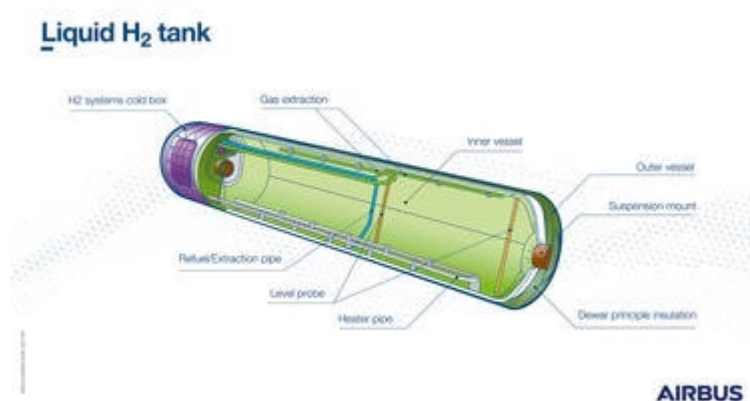
The main downside for Monel-400 however is the cost of it when compared to SA-516 and 316-Stainless Steel. Monel-400 costs significantly more when compared to the other materials however the difference in price can be justified by the difference in the lifespan of the material under the conditions of the Caustic Soda.

A Fuel Tank for a Liquid Hydrogen Engine

Component Design & Function

A fuel tank for a liquid hydrogen engine provides the required energy in the form of liquid for the engine to run, the tank must withstand the temperature and pressure of the hydrogen and prevent hydrogen embrittlement.

Figure 7, Liquid Hydrogen Tank Diagram



(Airbus 2025)

It is mostly cylindrical as the pressure in a tank is supposed to be spread evenly. However, because the liquid hydrogen's pressure is not as high, it is possible to make it more of an oval shape for better storage as demonstrated by Toyota's TG-RR corolla race

car in 2026, to add more, they have put the pump motor within the tank as well to get a superconducted motor aiming for more efficiency.

The use life of it will have to be over 10 years, as this is when most cars retire. The tank must last over 10 years as it must store the liquid hydrogen safely before it is taken to recycle. If it fails and breaks before that, liquid hydrogen is at very low temperature and significantly higher pressure than the atmosphere, making it dangerous if the hydrogen is released uncontrolled.

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Operating Environment

Table 22, Liquid Hydrogen Tank Operating Environment

Factor	Value/ Description	Additional Info
Service Temp	20K	Temp hydrogen boils at (1atm)
Operating Pressure	<0.85 Mpa	1 atm pressure operating
Cyclic Loading	>240000	NASA report range
Stress range	172.37-234.42 Mpa	NASA report range
Chemical Environment	LH2+GH2	NASA report range
Volumetric Energy Density	8 MJ/L	NASA report range
Failure modes	Fibre breakage, Stress corrosion, linear collapse, brittle fracture, boil-off fatigue	Failure assessments
Service Life	10-25 years	industry averages
Mass	High Importance	mass centric applications (aero)
Manufacturing	Plates (metal), fibre construction (composite)	Industry

Required Mechanical and Physical Properties

Table 23, Liquid Hydrogen Tank Required Mechanical and Physical Properties

Required Property	Quantitative Target	Reason
Brittle fracture resistance (KO)	<20K	NASA Report
Fracture toughness (KO)	42.20 (metals), 1.98-6.04 (CFRP)	NASA Report
hydrogen embrittlement resistance (KO)	not embrittle after exposure to gaseous hydrogen (exceeds UTS of 1100 Mpa (metals))	NASA Report
Fatigue life	>240000 cycles	NASA Report
Density	Lowest Possible	Minimum weight impacts effectiveness
Cost per kg	Lowest Possible	Product cost
UTS	Highest possible	governs hoop stress
Impact Strength	Highest possible	Withstand potential damages
Leak before burst (KO)	Must be able to leak before breaking	NASA Report

Candidate Materials

Carbon Fibre Composite T300

This material has the lowest density out of the three at 1.76kg/m^3 , is immune to embrittlement due it being a resin, highest ultimate tensile strength and yield strength at 1916Mpa and 1654Mpa

But is not recommended as it has the highest cost at 55.55 AUD/kg, low fracture toughness of 1.98, lowest impact resistance at 6.4, and most importantly does not meet the safety requirement of Leak before burst, CFRP is mostly brittle and once cracked the hole will be enough for a burst and not a leak.

AA5083-O and 316L Stainless Steel

For both, they are able to support temperatures lower than 20K(Anannya Power Pvt Ltd 2026) (AZoM 2001), with steel overall having much higher impact resistance, young's modulus and fracture toughness, significantly lower cost.

Aluminium on the other hand has a much lower density that can result in a much lower weight, a better yield strength and ultimate tensile strength, a higher resistance to hydrogen embrittlement.

Overall, aluminium tends to have the better characteristics even if the cost is higher than that of steel

Tabulated Material Properties

Table 24, Liquid Hydrogen Tank Tabulated Material Properties

Property	Units	Al 2050-T84	CFRP Composite	316L Stainless Steel
Density	g/cm ³	0.098	1.58	8.03
Young's Modulus	GPa	82.05 (77.59K)	131 (218.15K)	193
Yield Strength	MPa	527.45 (77.59K compressive)	1654.74 (218.15K compressive)	255.11
UTS	MPa	612.25 (77.59K)	1916.74 (218.15K)	530.9
Fracture Toughness	MPa·m ^{0.5}	48.79 (77.59K)	1.98 (297.04K)	68 (4K)
Cost	AUD/kg	44.02	55.55	3.98
High Velocity Impact Resistance	J	37.3	6.4	157.7
Hydrogen Embrittlement Resistance	--	70	100	45

Derived from (Boddorff & Langley Research Center 2025b, fig. Table 1), ("12K 300g Unidirectional Prepreg Carbon Fiber Fabric with Epoxy Resin for Carbon Fibre Parts" n.d.), ("Bharat Aerospace Metals: Aluminium Alloy 2050" n.d.), ("Stainless Steel (316) Price Trend, Index and Forecast" n.d.), (AK Steel Corporation 2007), (Alloys n.d.), (American Bureau of Shipping 2025), (ASM Handbook, Volume 19: Fatigue and Fracture 1996, p. 1814), (Benzing 2023, sec. Abstract), (Boddorff & Langley Research Center 2025b, fig. Table 1, fig. Table 2, sec. Durability), (Hafley et al. 2016, fig. Table 14), (Li et al. 2026), (Xiao et al. 2025, fig. Table 6).

Normalised Weighted Decision Matrix

For UTS (MPa):

Equation 14, Liquid Hydrogen Tank Normalising UTS for AL 2050-T84

$$n_{AL\ 2050-T84} = \frac{\sigma_{raw}}{\sigma_{base}} = \frac{612.25}{612.25} = 1.00$$

Equation 15, Liquid Hydrogen Tank Normalising UTS for CFRP Composite

$$n_{CFRP\ Composite} = \frac{\sigma_{raw}}{\sigma_{base}} = \frac{1916.74}{612.25} = 3.13 \rightarrow 3.00$$

Equation 16, Liquid Hydrogen Tank Normalising UTS for 316L

$$n_{316L} = \frac{\sigma_{raw}}{\sigma_{base}} = \frac{530.9}{612.25} = 0.87$$

Table 25, Liquid Hydrogen Tank Weight Allocation

Rank	Property	Weighting	Justification
5	Impact Strength	0.10	Not become brittle at operating temps
1	Fracture toughness	0.29	fail leak before break requirements
2	hydrogen embrittlement resistance	0.24	Not become brittle due to gaseous hydrogen
4	Density	0.14	major factor for weight of tank
6	Cost per kg	0.05	most inexpensive product
3	UTS	0.19	impacts hoop stress
	Σ	1.00	

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Table 26, Liquid Hydrogen Tank Normalised Material Properties

Property	Baseline	AL 2050-T84 raw	AL 2050-T84 norm	CFRP Composite raw	CFRP Composite norm	316L raw	316L norm
Impact Strength	37.3	37.3	1.00	6.4	0.17	157.7	3.00
Fracture toughness	48.79	48.79	1.00	1.98	0.04	68	1.39
hydrogen embrittlement resistance	70	70	1.00	100	1.43	45	0.64
Density	0.098	0.098	1.00	1.58	0.06	7.9	0.01
Cost per kg	44.02	44.02	1.00	55.55	0.79	3.98	3.00
UTS	612.25	612.25	1.00	1916.74	3.00	530.9	0.87
(colour means capped at 3)							

Table 27, Liquid Hydrogen Tank Weighted Decision Matrix

Property	Weight	AL 2050-T84 norm	AL 2050-T84×W	CFRP Composite norm	CFRP Composite ×W	316L norm	316L×W
Impact Strength	0.10	1.00	0.0952	0.17	0.0163	3.00	0.2857
Fracture toughness	0.29	1.00	0.2857	0.04	0.0116	1.39	0.3982
hydrogen embrittlement resistance	0.24	1.00	0.2381	1.43	0.3401	0.64	0.1531
Density	0.14	1.00	0.1429	0.06	0.0089	0.01	0.0018
Cost per kg	0.05	1.00	0.0476	0.79	0.0377	3.00	0.1429
UTS	0.19	1.00	0.1905	3.00	0.5714	0.87	0.1652
Σ			1.000		0.986		1.147

Table 28, Liquid Hydrogen Tank Final Material Selection

Rank	Material	Σ	Knock out check
1	316L Stainless Steel	1.147	✓
2	Al 2050-T84	1.000	✓
3	CFRP Composite	0.986	✓

Final Material Selection

Aluminium is final choice made, as for a car it should be lightweight otherwise it will greatly increase the weight of the car, it will also experience a lower centre of gravity as the tank is usually very big and will take up a lot of space on top of the original centre of gravity. For the structure itself, it is beneficial as having a lower density means more materials can be used to thicken up the walls without impacting the weight as much. Moreover, thicker material and higher resistance to liquid hydrogen also means that it will last a lot longer than if it was made of steel coupled with the resistance to embrittlement for the 5000series aluminium being used. It also meets the requirement for having leaks before cracking for safety.

However, on the disadvantages, aluminium is over 10 times more expensive than steel. The higher heat conductivity for aluminium will also lead to warmer air in the atmosphere to creep in the tank more easily and lead to boil off. It also has a much lower impact resistance and overall strength compared to steel, which they may be more protection or damper systems to prevent the aluminium tank to take too much shock. Manufacturing will also be harder as welding aluminium requires special equipment unlike steel which is simpler.

Turbine Blade for a Turbojet Engine

Component Design & Function

The component, a high-pressure turbine blade, is used in a very high temperature section of a turbojet engine. The core function that the blade does, is extracting the required energy from the elevated temperature and pressure combustion gases which allows for the conversion process to occur; from gas energy that translates into rotational motion through the turbine shaft. The shaft continues in a rotation motion which thus drives the compressor and allows for the air to compress and for combustion occur within the engine. The blade operates in a severe condition, as the environment surrounding the engine consists of thermal loading, centrifugal stress, gas

bending, vibration, oxidation, hot corrosion, and fatigue.

The turbine blade is critical for safety, as it is a core structural component for the turbojet engine. The rotating air foil section is designed to maintain required energy from the gas glow, as the root of the blade is mechanically locked into the turbine disk which allows centrifugal loads to be transferred during maximised speed rotation. Due to high metal temperatures during the process, the blade must also inhibit cooling attributes and surface film-cooling to ensure stable temperatures. Another common addition is protective thermal barrier coatings that decrease the heat transfer into the metal which causes the metal to resist against oxidation and therefore corrosion. Thus, the material selected must be able to provide extreme temperatures strength, creep, fatigue, oxidation/corrosion resistance, etc.

Figure 8, (Tungaloy Corporation 2023)



(Tungaloy Corporation 2023)

The main function of a turbine blade is to extract energy from combustion gases and keep the compressor running by spinning the shaft connected to the rotor. Because of these tough conditions, the material must be able to withstand under high temperature and pressurized situations. The turbine blade use in aircrafts requires safety under top priority as any potential fault risks the lives of occupants onboard the aircraft.

The gas temperatures reach exceedingly high temperatures while the blade undergoes high rotational speeds over long flight cycles which the material must avoid total failure to avoid a catastrophic accident when used on an aircraft. Cost is generally ignored in favour of maximised safety if requirements are met.

Operating Environment

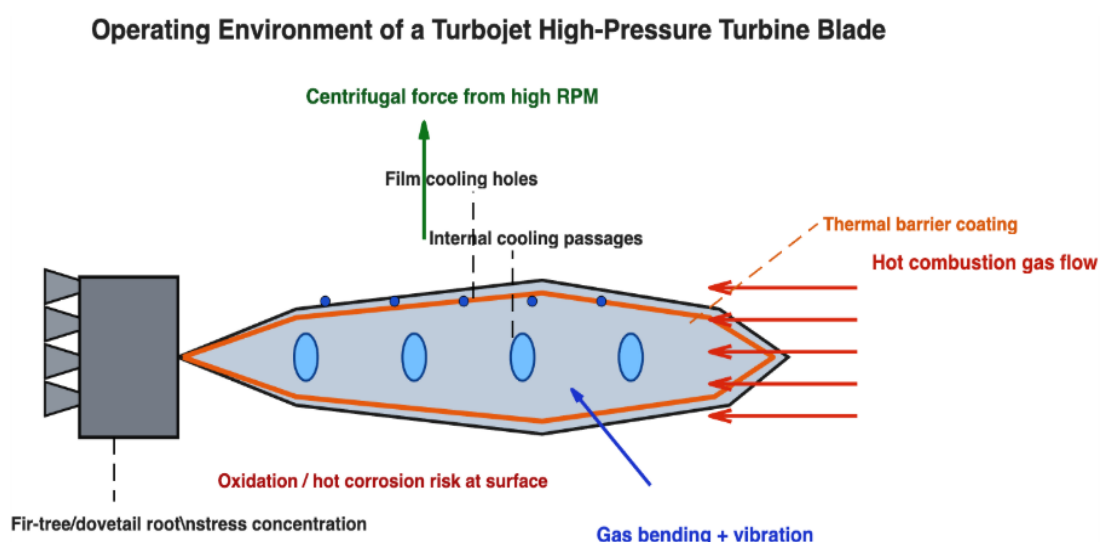
Table 29, Turbine Blade Operating Environment

Factor	Value/ Description	Additional Info
Combustors exit gas temp	1700C	From report
Gas temp range	900C-1700C	Report
Internal cooling	operating temps can reach far above the max material temp, so can be cooled	Report
Service Temp	>700C	report
Rotational speed	>10000 RPM	report
Cyclic loading	fatigue from high frequency of rotations	report
Thermal fatigue	Changes in temp between operating conditions and ambient conditions	report
Surface protection	Thermal coating to increase melting temp	Report
Failure modes	Creep, corrosion, thermal fatigue	Report
Service Life	>25000 flight cycles	Longevity to save having to change blades
Cost	Minimum cost for maximum reward from material	Save money whilst still having reliable blade

The specific operating environment for a turbojet high pressure turbine blade is formed by the effects of high temperature, the increased rotational speed alongside the extreme combustion gases. The combustion gas temperatures can be significantly higher than the safe operating temperature of most metallic materials/alloys, which indicates that the blade material is unable to be substituted for room temperature strength. Another significant factor is the blade's retained strength at extreme temperature, it is also resistant to creep deformation, and its ability to resist repeated thermal and rotational mechanical cycling. The operating environment table therefore identifies major factors such as the following: service temperature, rotational loading, cyclic fatigue, thermal fatigue, oxidation, hot corrosion, and surface protection.

Creep is also treated as a major environmental factor because the blade undergoes sustained centrifugal stress while also operating at high temperature. With these conditions, the material can begin to slowly deform even in the case that the applied stress is significantly below its short-term yield strength. There has been research on directionally solidified nickel-based turbine blade superalloys which demonstrate that increasing turbine speed and the inlet temperature increase the mechanical loading and can cause creep fracture; one of the major failure modes in aero-engine turbine blades (Shi et al., 2025). This thus proves the need to include creep rupture strength and elevated-temperature yield strength as critical requirements for the device.

Figure 9, Turbojet Turbine Blade Original Operating Environment Original Diagram



In addition, there are other concerns such as fatigue and thermal fatigue. During engine start-up, shutdown and throttle changes, the blade repeatedly heats and cools, which causes expansion and contraction. These conditions can be the cause of low and

high cycle fatigue and crack growth. Therefore, fatigue strength, fracture toughness and thermal fatigue resistance are also necessary requirements.

The chemical environment is also extreme because the blade surface is exposed to high temperature combustion gases. The blade surface can oxidate and thus degrade the surface of the blade, and hot corrosion can only cause damage to the blade surface as well. Thus oxidation/hot-corrosion resistance and coating compatibility must be included in the material selection process.

Required Mechanical and Physical Properties

Table 30, Turbine Blade Required Mechanical and Physical Properties

Required Property	Quantitative Target	Reason
Service Temp (KO)	>700	Withstand environment
Low CTE	Lowest possible	Keep strain and thermal stress low
Density	Lowest possible	minimise power and weight
Fatigue Strength	Highest possible	Withstand cycles
Youngs Modulus	Highest possible	withstand stresses
UTS	Highest possible	withstand stresses
Corrosion	Lowest possible	Keep dimensions the same
Creep rupture Stress	Lowest possible	Avoid failure modes
Yield Strength	Highest possible	Avoid failure modes
Cost	lowest possible	Minimise product price

As discussed previously in the operating environment, there are necessary mechanical and physical properties for the turbine blade operating environment.

Since the blade works at very high temperature and is also operating while carrying centrifugal and gas-bending loads, elevated-temperature properties required for the mechanism. There is also no need for service temperature capability as a material that is unable to survive the hot environment is unsuitable even if it has factors like density, cost, or room-temperature strength. This is why Ti-6Al-2Sn-4Zr-2Mo can be included as a comparison material but is not suitable in the case that its service temperature limit is below the required range for operating conditions.

Moving on to creep rupture strength, it is one of those important properties for this application as a turbine blade is likely to undergo long periods of extreme stress at very high temperatures, this usually occurs during engine operation in which the materials must have the capability to resist creep deformation alongside rupture in the long-term usage. Yield strength and ultimate tensile strength should also be considered at elevated temperatures since the blade must resist permanent deformation and fracture.

Density is included because it directly affects centrifugal loading. However, density is not a factor that can be prioritised over the service temperature and creep performance. A lightweight material that cannot retain strength at the required turbine temperature is unsafe.

Candidate Materials

René 41, UDIMET 700 and Ti-6Al-2Sn-4Zr-2Mo were selected to compare two nickel-based superalloys with one lightweight aerospace titanium alloy. Nickel-based superalloys are the most important considerations because high-pressure turbine blades require high-temperature strength, creep resistance, fatigue resistance, and oxidation/hot-corrosion resistance. Literature shows that turbine blade materials have developed from wrought alloys to cast, directionally solidified and single-crystal alloys as turbine entry temperatures increased (Yadav et al., 2020).

René 41 and UDIMET 700 were included because they are high-temperature nickel-based superalloys with useful qualities such as elevated-temperature strength and turbine-related service capability. Ti-6Al-2Sn-4Zr-2Mo was included as a lightweight comparison material. Although its low density reduces the centrifugal loading, its lower service temperature makes it incompatible.

Tabulated Material Properties

Table 31, Turbine Blade Tabulated Material Properties

Property	Units	Rene 41	UDIMET 700	Ti-6Al-2Sn-4Zr-2Mo
Density	g/cm ³	8.249	8.25	4.54
Young's Modulus	GPa	152 (982 C)	132	118
Yield Strength	MPa	345 (925 C)	23	1050
UTS	MPa	400 (925 C)	115-234	1110
Elongation at Break	%	26 (925)	44	13
Fatigue Strength	MPa	138	255.1	280
CTE	μm/m·°C	17.8 (1093.33 C)	17.5	8.1
Melting Point	°C	1316-1371	1177	1705
Specific Heat	J/kg·K	0.452	435	460
Service Temp Limit	°C	982	1050	540
Corrosion Rate	mm/yr	7.008	14.02	0.0876
Cost	AUD/kg	76.95	90.94	44.19
Stress Rupture	--	(10h: 189MPa) (200h: 83MPa) 925C	(10h: 117Mpa) (1000h: 55.2MPa) 982C	--
Creep Rupture	--	100h: 128MPa - 815C	193h: 103MPa 982 C	100h: 170 Mpa

Derived from ("Rene 41 Tech data" n.d.), ("high temperature nickel superalloys 1 | Total Materia" n.d., sec. Introduction to Nickel-Based Superalloys), ("Titanium 6Al 2Sn 4Zr 2Mo" n.d.), ("Titanium Ti-6Al-2Sn-4Zr-2Mo (Ti-6-2-4-2), Duplex Annealed" n.d.), ("Titanium Ti-6Al-2Sn-4Zr-2Mo-0.1Si; Duplex Annealed" n.d.), ("UDIMET 630 UDIMET 700 UDIMET 710 UNITEMP AF2-1DA | Products for sale | Sell Metal and Steel (#54810)" n.d.), ("UDIMET 700 and ALSL 687 chemical composition and density 8.25gm/cm³" n.d.), (AZoM 2023), (Boesch & Canada 1968), (Conn, Rudy, & Hydronautics, Inc. 1972, fig. Table 1), (Fargas et al. 2021, sec. 3), (Galati et al. 2021, sec. Introduction), (Meta-Author n.d.), (Phillips, NASA, & Langley Research Center 1965, fig. Table III), (SAE J 467 1968), (Smiths Advanced Metals 2022), (Superalloys 2019), (Titanium Metals Corporation n.d., fig. Table 7), (Wilson 1974, fig. Table II), (Yiyunyingliping 2024).

The data should be interpreted carefully because turbine blade performance depends on temperature, heat treatment, alongside the exposure time and the testing environment. Elevated-temperature data is more relevant than room-temperature data, especially for yield strength, UTS, creep rupture stress, fatigue strength, and corrosion resistance.

The values in the table are important however do not consider the fact that properties such as hot corrosion rate and creep rupture strength are usually not measured under the same conditions. Thus, the material comparison is good for ranking, but the final

decision should still consider the knockout requirements like service temperature and creep resistance.

Normalised Weighted Decision Matrix

For Density (g/cm³):

Equation 17, Turbine Blade Normalising Density for Rene 41

$$n_{\text{Rene 41}} = \frac{\rho_{\text{base}}}{\rho_{\text{raw}}} = \frac{8.249}{8.249} = 1.00$$

Equation 18, Turbine Blade Normalising Density for UDIMET 700

$$n_{\text{UDIMET 700}} = \frac{\rho_{\text{base}}}{\rho_{\text{raw}}} = \frac{8.249}{8.25} = 1.00$$

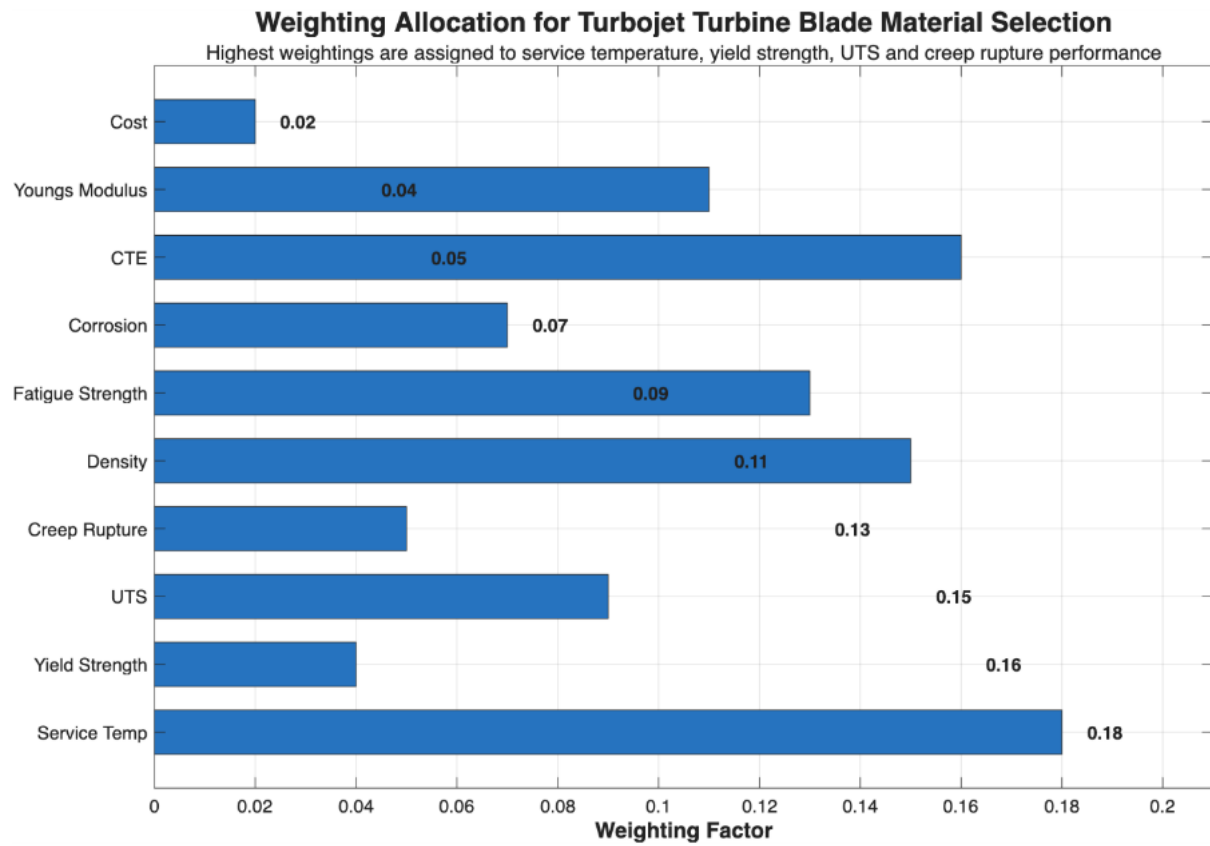
Equation 19, Turbine Blade Normalising Density for Ti-6Al-2Sn-4Zr-2Mo

$$n_{\text{Ti-6Al-2Sn-4Zr-2Mo}} = \frac{\rho_{\text{base}}}{\rho_{\text{raw}}} = \frac{8.249}{4.54} = 1.82$$

Table 32, Turbine Blade Weight Allocation

Rank	Property	Weighting	Justification
1	Service Temp	0.18	Withstand environment
8	CTE	0.05	Keep strain and thermal stress low
5	Density	0.11	minimise power and weight
6	Fatigue Strength	0.09	Withstand cycles
9	Youngs Modulus	0.04	withstand stresses
3	UTS	0.15	withstand stresses
7	Corrosion	0.07	Keep dimensions the same
4	Creep rupture Stress	0.13	Avoid failure modes
2	Yield Strength	0.16	Avoid failure modes
10	Cost	0.02	Minimise product price
	Σ	1.00	

Figure 10, Turbine Blade Weight Allocation Graph



Continue to next page

Table 33, Turbine Blade Normalised Material Properties

Property	Baseline	Rene 41 raw	Rene 41 norm	UDIMET 700 raw	UDIMET 700 norm	Ti-6Al- 2Sn- 4Zr-2Mo raw	Ti-6Al- 2Sn- 4Zr-2Mo norm
Service Temp	982	982	1.00	1050	1.07	540	0.00
CTE	17.8	17.8	1.00	17.5	1.02	8.1	2.20
Density	8.249	8.249	1.00	8.25	1.00	4.54	1.82
Fatigue Strength	138	138	1.00	255.1	1.85	280	2.03
Youngs Modulus	152	152	1.00	132	0.87	118	0.78
UTS	400	400	1.00	115	0.29	1110	2.78
Corrosion	7.008	7.008	1.00	14.02	0.50	0.0876	5.00
Creep rupture Stress	128	128	1.00	103	1.24	170	0.75
Yield Strength	345	345	1.00	23	0.07	1050	3.00
Cost	76.95	76.95	1.00	90.94	0.85	44.19	1.74
(colour means capped at 3)							

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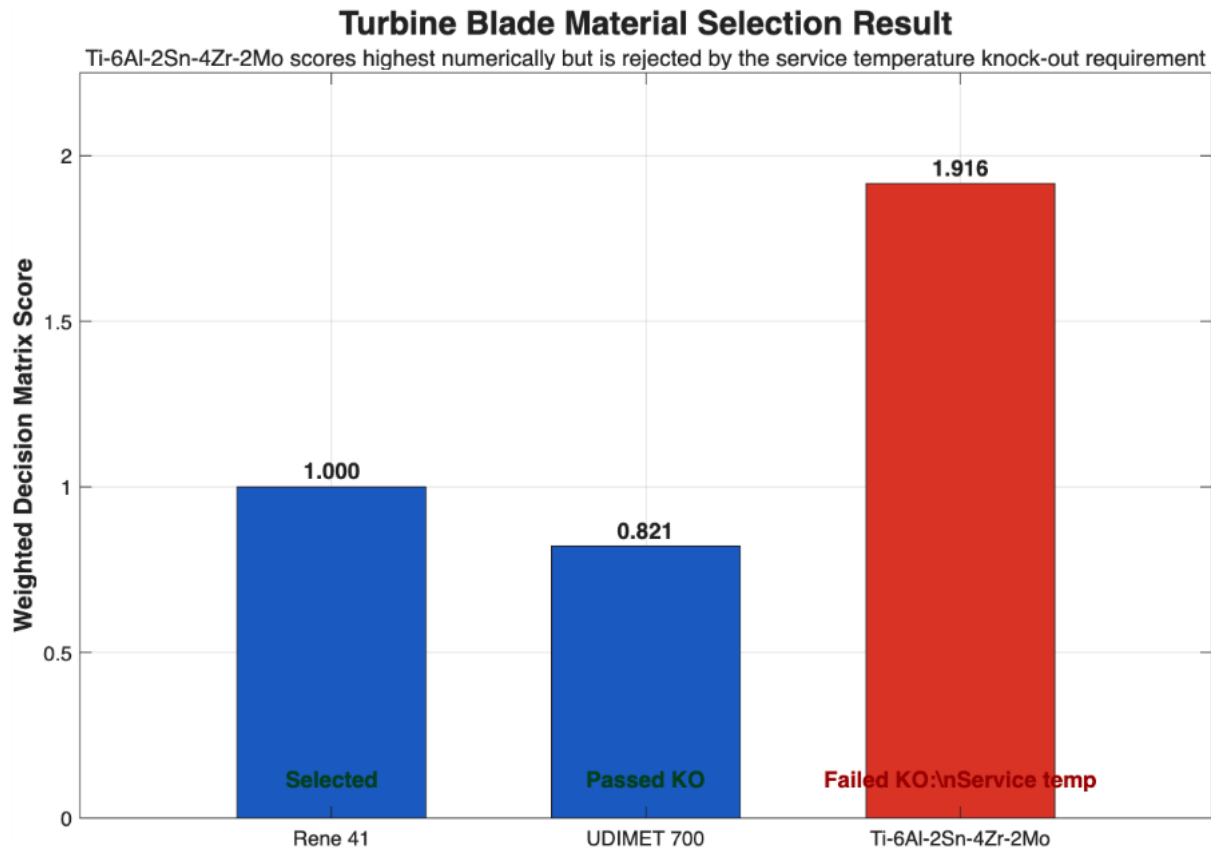
Table 34, Turbine Blade Weighted Decision Matrix

Property	Weight	Rene 41 norm	Rene 41×W	UDIMET 700 norm	UDIMET 700×W	Ti-6Al-2Sn-4Zr-2Mo norm	Ti-6Al-2Sn-4Zr-2Mo×W
Service Temp	0.18	1.00	0.1818	1.07	0.1944	0.00	0.0000
CTE	0.05	1.00	0.0545	1.02	0.0555	2.20	0.1199
Density	0.11	1.00	0.1091	1.00	0.1091	1.82	0.1982
Fatigue Strength	0.09	1.00	0.0909	1.85	0.1681	2.03	0.1845
Youngs Modulus	0.04	1.00	0.0364	0.87	0.0316	0.78	0.0282
UTS	0.15	1.00	0.1455	0.29	0.0418	2.78	0.4036
Corrosion	0.07	1.00	0.0727	0.50	0.0364	5.00	0.3636
Creep rupture Stress	0.13	1.00	0.1273	1.24	0.1582	0.75	0.0958
Yield Strength	0.16	1.00	0.1636	0.07	0.0109	3.00	0.4909
Cost	0.02	1.00	0.0182	0.85	0.0154	1.74	0.0317
Σ			1.000		0.821		1.916

Table 35, Turbine Blade Final Material Selection

Rank	Material	Σ	Knock out check
1	Rene 41	1.000	✓
2	UDIMET 700	0.821	✓
3	Ti-6Al-2Sn-4Zr-2Mo	1.916	X-Doesn't satisfy minimum service temperature

Figure 11, Turbine Blade Weighted Decision Matrix Result Graph



The weighting system reflects the main demands of a turbojet turbine blade. Service temperature, yield strength, UTS and creep rupture strength receive high weightings because they control whether the material can withstand the hot-section environment. Density, fatigue strength, CTE and corrosion resistance are also important because they can affect centrifugal stress, cyclic durability, thermal fatigue, and long-term surface damage.

Ti-6Al-2Sn-4Zr-2Mo achieves the highest score because it is lightweight and strong, but it fails the service-temperature requirement. Therefore, it cannot be selected even though it has a high score.

Final Material Selection

René 41 was selected as the most suitable material because it satisfies the following conditions: service-temperature requirement and provides the best overall balance of elevated-temperature strength, creep rupture performance, oxidation/corrosion resistance, and manufacturability. UDIMET 700 is also a suitable nickel-based superalloy, but René 41 ranked higher after the weighted comparison and the knockout check.

Ti-6Al-2Sn-4Zr-2Mo was rejected because it does not meet the temperature requirement, even though it scored highly in the matrix. It does possess low density, but for a high-pressure turbine blade, high-temperature capability and creep resistance are more important than the weight or cost. The selected René 41 blade should also use cooling features and a thermal barrier coating to improve service life in the hot-section environment.

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Comparative Discussion

For the bumper bar, DP590 Dual phase Steel was chosen as it matches the requirements the most out of all the selections, however, has trade-offs of lower recyclability and lifetime span. The steel was the strongest and most durable material, which is required in maximising the safety of occupants of the vehicle rather than a cheaper alternative.

For the window latch, AA6063 Anodised was chosen over C95800 due its favourable properties in manufacturing for a high supply item where cost is minimal. Its lower carbon footprint may also be wanted in comparison for a stronger latch that is harder to produce.

Monel 400 was chosen for the caustic soda vessel as other materials lack the caustic resistance required for long term storage, although it costs significantly more to manufacture. The use of weaker materials may compromise the safety of application users and potentially damage the surrounding environment if its internal contents were exposed.

Aluminium (Al 2050-T84) was chosen for the liquid hydrogen tank despite receiving a lower score than stainless steel as it is lightweight and easier to transport which other selections struggle to match, however is much harder to manufacture and costs more. The metal also lasts significantly longer, decreasing the chance of an accidental exposure to the area proximity.

Rene 41 was chosen over UDIMET 700 for a turbine blade as it has significantly higher strength under extreme conditions which essential for occupants on the turbojet while matching the requirements for standard use. The material is also cheaper to manufacture, allowing for more economical use on different applications without compromising safety of passengers when used on aircrafts.

Conclusion

The final selection of material for each application was chosen based on the required properties unique circumstances and potential situations regardless of whether it was the top of the weighted score. Material properties were also carefully determined if they matched all the requirements for the application and were avoided for selection if deemed a risk for use. Data from sources was carefully examined individually and researched widely to reduce bias and possibility of incorrect information.

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Appendix D – AI Usage Form

AI Declaration Form

Course: MIET2115 – Mechanics and Materials 2

Team (Individual) Name / Team (Student) Number:

1. Declaration of AI Use

AI tools include (but are not limited to): VAL, ChatGPT, Claude, GitHub Copilot, Grammarly, DeepSeek, Bing/Google AI features, or any software/online service that generates or edits text, code, images, or design outputs.

Please select one option below:

- ☐ **No AI Use:** We did not use any AI tools in preparing this report.
- ☐ **Language/Editing Support Only:** We/I used AI tools only for grammar, spelling, punctuation, formatting, or general language clarity. No technical, analytical, or design content was generated using AI.
- ☒ **AI Used for Understanding or Early Drafting:** We/I used AI tools to assist understanding, brainstorm ideas, or generate preliminary drafts (e.g., explanations, pseudocode, design concepts). All AI-assisted content was independently checked, verified, and rewritten in our own words.
- ☒ **Other AI Use:** We/I used AI tools in other ways not covered above. Details are provided below.

Used to find a datapoint and location of said info from a supplied source if available from source.

2. Description of AI Use (if applicable)

If any AI use was selected above, briefly describe what was used and how it contributed to your work (e.g., clarifying theory, debugging hints, summarising articles, formatting tables, generating images, refining explanations, etc.).

Example: “We used ChatGPT to rephrase parts of the static analysis section for clarity. All calculations, interpretations, and final explanations were completed by us.”

Used to figure out what headings should be used in the report.
Used to find a datapoint and location of said info from a supplied source if available from source.

3. Verification Statement

By signing this form, we/I confirm that:

- All technical analysis, calculations, design decisions, and interpretations in the submission are our/my own work and based on our/my own understanding.
- Any AI-generated ideas or suggestions have been critically evaluated, verified, and edited by us/me.
- No AI tool was used to produce final technical results, code, or design outputs unless explicitly permitted in the assessment instructions.
- This submission complies with the academic integrity requirements of this course and the University.
- Where there are doubts as to the originality of this assessment, the course coordinator may interview the team to determine the learning outcomes.

Signature



Team Representative / Individual Student Name: Hugh Williams

Signature: ^^^^^^^^^^^^

Date: 09/06/2026

Note: For group submissions, the representative signs on behalf of the group and is responsible for ensuring all members agree with the declaration.